

Itô Calculus

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First, an example

Before the formal machinery, compute $d(W_t^2)$ where W_t is standard Brownian motion. Naive calculus would give $2W_t dW_t$, copying $\frac{d}{dx}x^2 = 2x$. But Brownian motion has nonzero quadratic variation, so Itô's lemma adds a correction:

$$d(W_t^2) = 2W_t dW_t + dt.$$

Read aloud: “d of W sub t squared equals two W sub t d W sub t plus d t.” The extra dt comes from the heuristic $(dW_t)^2 = dt$ — the quadratic variation of Brownian motion. Equivalently, in integral form, $W_1^2 - W_0^2 = 2 \int_0^1 W_s dW_s + 1$. Since the Itô integral is a martingale with mean zero, taking expectations recovers the familiar $\mathbb{E}[W_1^2] = 1$. The correction term dt is exactly the bookkeeping needed to keep this identity consistent.

1 Setting

Fix a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$ satisfying the usual conditions and supporting a standard Brownian motion $W = (W_t)_{t \geq 0}$ adapted to $(\mathcal{F}_t)$. Sample paths of W are almost surely continuous and nowhere differentiable, with unbounded total variation on every compact interval.

Definition 1 (Quadratic variation). *For a stochastic process $X = (X_t)$ and a partition $\Pi = \{0 = t_0 < t_1 < \dots < t_n = t\}$ write $\|\Pi\| = \max_i(t_i - t_{i-1})$. The quadratic variation of X on $[0, t]$ is*

$$[X]_t := \lim_{\|\Pi\| \rightarrow 0} \sum_{i=1}^n (X_{t_i} - X_{t_{i-1}})^2,$$

Read aloud: “bracket X bracket sub t is defined as the limit, as the mesh of Pi goes to zero, of the sum from i equals one to n of X sub t-i minus X sub t-i-minus-one, all squared.” provided the limit exists in probability. For Brownian motion, $[W]_t = t$ a.s.

Definition 2 (Itô integral). *Let $H = (H_t)$ be a progressively measurable process with $\mathbb{E} \int_0^T H_s^2 ds < \infty$. The Itô integral of H against W is the L^2 -limit*

$$\int_0^t H_s dW_s := \lim_{\|\Pi\| \rightarrow 0} \sum_{i=1}^n H_{t_{i-1}} (W_{t_i} - W_{t_{i-1}}).$$

Read aloud: “the integral from zero to t of H sub s d W sub s is defined as the limit, as the mesh of Pi tends to zero, of the sum from i equals one to n of H sub t-i-minus-one times the quantity W sub t-i minus W sub t-i-minus-one.” Evaluation at the left endpoint t_{i-1} is essential: it makes $t \mapsto \int_0^t H dW$ a continuous square-integrable martingale, with Itô isometry $\mathbb{E}[(\int_0^t H dW)^2] = \mathbb{E} \int_0^t H_s^2 ds$.

Definition 3 (Itô process). A process X is an Itô process if

$$X_t = X_0 + \int_0^t b_s ds + \int_0^t \sigma_s dW_s,$$

Read aloud: “ X sub t equals X sub zero plus the integral from zero to t of b sub s ds plus the integral from zero to t of σ sub s dW sub s .” for adapted b, σ with $\int_0^T |b_s| ds < \infty$ and $\int_0^T \sigma_s^2 ds < \infty$ a.s. We write this symbolically as $dX_t = b_t dt + \sigma_t dW_t$.

2 Itô’s Lemma

Theorem 4 (Itô’s lemma, 1D). Let $X_t = X_0 + \int_0^t b_s ds + \int_0^t \sigma_s dW_s$ be an Itô process and let $f \in C^2(\mathbb{R})$. Then $f(X_t)$ is again an Itô process and

$$df(X_t) = f'(X_t) dX_t + \frac{1}{2} f''(X_t) \sigma_t^2 dt.$$

Read aloud: “ d of f of X sub t equals f prime of X sub t times dX sub t plus one half f double-prime of X sub t times σ sub t squared dt .” Equivalently, in integral form,

$$f(X_t) - f(X_0) = \int_0^t f'(X_s) b_s ds + \int_0^t f'(X_s) \sigma_s dW_s + \frac{1}{2} \int_0^t f''(X_s) \sigma_s^2 ds.$$

Read aloud: “ f of X sub t minus f of X sub zero equals the integral from zero to t of f prime of X sub s times b sub s ds , plus the integral from zero to t of f prime of X sub s times σ sub s dW sub s , plus one half times the integral from zero to t of f double-prime of X sub s times σ sub s squared ds .”

Proof sketch via second-order Taylor expansion and $(dW_t)^2 = dt$. Fix a partition $\Pi = \{0 = t_0 < \dots < t_n = t\}$ and write $\Delta_i X = X_{t_i} - X_{t_{i-1}}$, $\Delta_i t = t_i - t_{i-1}$, $\Delta_i W = W_{t_i} - W_{t_{i-1}}$. By telescoping and a second-order Taylor expansion of f around $X_{t_{i-1}}$,

$$\begin{aligned} f(X_t) - f(X_0) &= \sum_{i=1}^n [f(X_{t_i}) - f(X_{t_{i-1}})] \\ &= \sum_i f'(X_{t_{i-1}}) \Delta_i X + \frac{1}{2} \sum_i f''(X_{t_{i-1}}) (\Delta_i X)^2 + R_\Pi, \end{aligned}$$

Read aloud: “ f of X sub t minus f of X sub zero equals the sum from i equals one to n of the bracket f of X sub t_i minus f of X sub t_{i-1} , which equals the sum over i of f prime of X sub t_{i-1} times $\Delta_i X$, plus one half times the sum over i of f double-prime of X sub t_{i-1} times $\Delta_i X$ squared, plus R sub Π .” where R_Π collects the higher-order Taylor remainder, controlled by the modulus of continuity of f'' on a compact set containing the path of X .

First-order term. Since $\Delta_i X = \int_{t_{i-1}}^{t_i} b_s ds + \int_{t_{i-1}}^{t_i} \sigma_s dW_s$, the sum $\sum_i f'(X_{t_{i-1}}) \Delta_i X$ converges in L^2 to

$$\int_0^t f'(X_s) b_s ds + \int_0^t f'(X_s) \sigma_s dW_s,$$

Read aloud: “the integral from zero to t of f prime of X sub s times b sub s ds , plus the integral from zero to t of f prime of X sub s times σ sub s dW sub s .” by definition of the Itô integral and

dominated convergence; the left-endpoint evaluation $f'(X_{t_{i-1}})$ is exactly what the Itô construction requires.

Second-order term — the key identity $(dW_t)^2 = dt$. Expand

$$(\Delta_i X)^2 = b_{t_{i-1}}^2 (\Delta_i t)^2 + 2b_{t_{i-1}} \sigma_{t_{i-1}} (\Delta_i t)(\Delta_i W) + \sigma_{t_{i-1}}^2 (\Delta_i W)^2 + o(\Delta_i t)\text{-corrections}.$$

Read aloud: “delta-i X squared equals b sub t-i-minus-one squared times delta-i t squared, plus two times b sub t-i-minus-one times sigma sub t-i-minus-one times delta-i t times delta-i W, plus sigma sub t-i-minus-one squared times delta-i W squared, plus little-o of delta-i t corrections.” The $(\Delta_i t)^2$ contributions sum to $O(\|\Pi\| \cdot t) \rightarrow 0$. The cross-terms $(\Delta_i t)(\Delta_i W)$ have L^2 -norm $O(\Delta_i t^{3/2})$, summing to $O(\|\Pi\|^{1/2}) \rightarrow 0$. The surviving piece is

$$\sum_i f''(X_{t_{i-1}}) \sigma_{t_{i-1}}^2 (\Delta_i W)^2.$$

Read aloud: “the sum over i of f double-prime of X sub t-i-minus-one times sigma sub t-i-minus-one squared times delta-i W squared.” Because $\mathbb{E}[(\Delta_i W)^2] = \Delta_i t$ and $\text{Var}[(\Delta_i W)^2] = 2(\Delta_i t)^2$, a direct L^2 computation shows that $(\Delta_i W)^2$ may be replaced by $\Delta_i t$ in the limit:

$$\sum_i f''(X_{t_{i-1}}) \sigma_{t_{i-1}}^2 (\Delta_i W)^2 \xrightarrow[\|\Pi\| \rightarrow 0]{L^2} \int_0^t f''(X_s) \sigma_s^2 ds.$$

Read aloud: “the sum over i of f double-prime of X sub t-i-minus-one times sigma sub t-i-minus-one squared times delta-i W squared converges in L-two, as the mesh of Pi tends to zero, to the integral from zero to t of f double-prime of X sub s times sigma sub s squared d s.” This is precisely the heuristic identity $(dW_t)^2 = dt$ made rigorous, and it is an instance of the quadratic variation $[W]_t = t$.

Remainder. $R_\Pi \rightarrow 0$ in probability because f'' is continuous and the path of X is bounded on $[0, t]$ almost surely. Collecting the surviving pieces yields the formula stated. $\square$

Remark 5 (Why the left endpoint?). *The proof shows the limit of $\sum H_{t_{i-1}} \Delta_i W$ depends on the choice of evaluation point. Replacing $H_{t_{i-1}}$ with the midpoint $H_{(t_{i-1}+t_i)/2}$ gives the Stratonovich integral, which obeys the classical chain rule $df(X_t) = f'(X_t) \circ dX_t$ but is not a martingale. The right-endpoint choice peeks at the future and produces a different, anticipating object. Itô’s left-endpoint choice gives a martingale, at the cost of the second-order correction term in Itô’s lemma.*

Remark 6 (Canonical example). *With $f(x) = x^2$ and $X_t = W_t$, Itô’s lemma yields*

$$d(W_t^2) = 2W_t dW_t + dt \implies \int_0^t W_s dW_s = \frac{1}{2}(W_t^2 - t).$$

Read aloud: “d of W sub t squared equals two W sub t d W sub t plus d t, which implies that the integral from zero to t of W sub s d W sub s equals one half of the quantity W sub t squared minus t.” The classical guess $\frac{1}{2}W_t^2$ is off by exactly the Itô correction $-t/2$.